

PINNGraPE

Physics-Informed Neural Network for Gravitational-wave Parameter Estimation

M. Scialpi¹ M. Bejger²

¹ Department of Physics and Earth Science, University of Ferrara, Italy

² INFN, Ferrara section, Italy

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Abstract

Detected Gravitational Waves (GWs) are goldmines of information on the compact binary emitting systems. Usually MCMC techniques infer parameter's values in a 15-dimensional parameter space in an accurate way, but they are very lengthy. On the other hand, Physics-Informed Neural Networks (PINNs) [3] are a rapidly emerging branch of Supervised Machine Learning, devoted precisely to solve physical problems. This poster will present the PINNGraPE algorithm, which shows how PINNs can help to speed up the inference process. This algorithm in particular is able to infer two combinations of BBH's masses to 10^{-2} relative accuracy, thanks to a 1.5PN formalism [1].

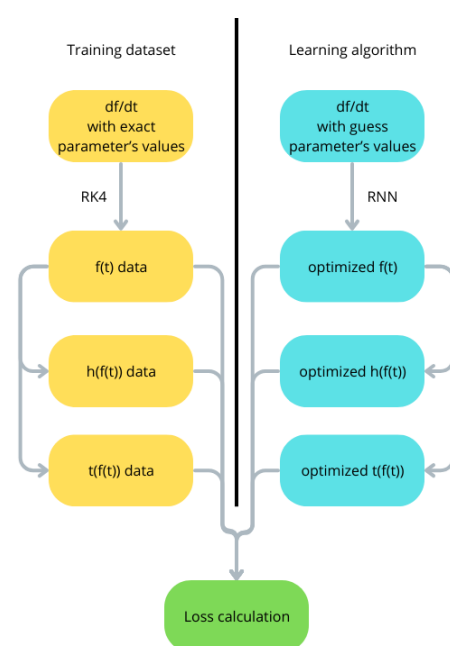
1 Introduction

- **Goal:** to infer η and M_{tot} values from a waveform data, with

$$M_{tot} = m_1 + m_2, \quad \eta = \frac{m_1 m_2}{M_{tot}^2} \quad (1)$$

and m_1 and m_2 the two BH's masses.

- **Training dataset:** a single simulated waveform $\{t_k, f_k, h(f_k)\}$, with target parameters: $M_{tot} = 60$, $\eta = 0.25$.
- PINNGraPE is based on a RNN with a RK4 integrator implemented inside the recurrent cell [2].



PINNGraPE workflow.

2 Methods: PINNGraPE

We impose

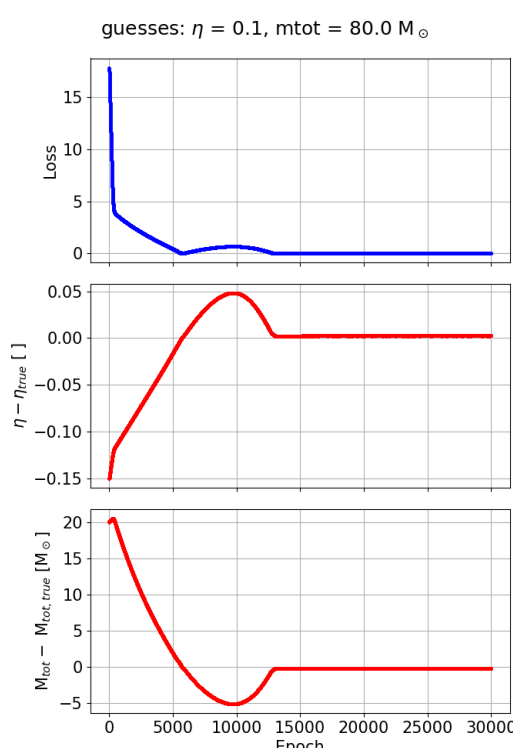
$$\frac{df}{dt} = F[f, \eta, M_{tot}], \quad (2)$$

where F is the 1.5PN frequency evolution functional. PINNGraPE solves (2) thanks to the RNN and calculates $h(f)$ thanks to 1.5PN formalism. It optimizes $\{\eta, M_{tot}\}$ parameters to minimize the loss function

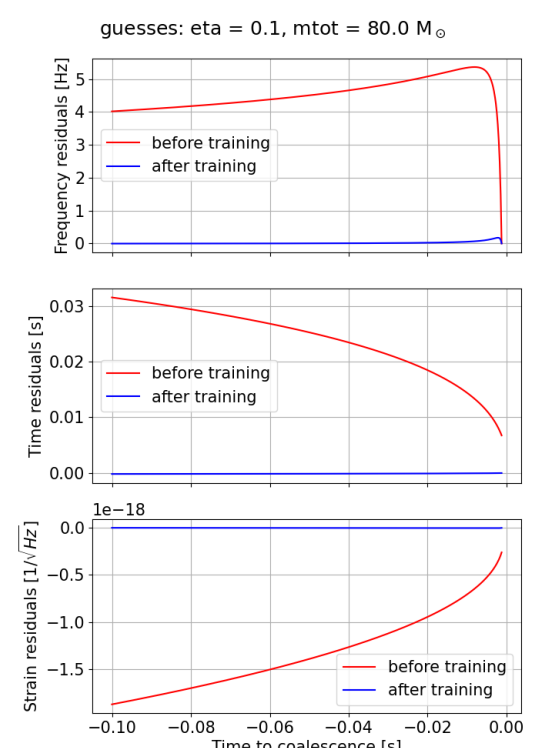
$$L = \frac{\beta_f}{N} \sum_{k=1}^N |f_k - f(t_k)| + \frac{\beta_h}{N} \sum_{k=1}^N |h_k - h(f(t_k))|, \quad (3)$$

where $\beta_f = 1, \beta_h = 10^{19}$

3 Results



- On the left plots you can see the loss behaviour and the two parameter optimization wrt epochs. The jump is due to a combination of the loss function behaviour wrt the parameters and the AdamW optimizer operation.
- On the right plots you can see the residuals for $f(t_k)$, for $t(f(t_k))$ (just for reference) and for $h(t_k)$.
- Guess parameters: $\eta = 0.1000$, $M_{tot} = 80.00$
- Guess parameters: $\eta = 0.2524$, $M_{tot} = 59.74$
- Final loss value: 2.437×10^{-2} .



4 Discussion

Main result: PINNGraPE is able to infer both η and M_{tot} with a relative error of 10^{-2} .

Next steps:

- to build a real dataset spanning a physical parameter space;
- to test robustness against noise and glitches;
- to extend the number of parameters to infer.

(Not so) **Remote steps:**

- use of cWB real outputs,
- apply PINNs approach to TOV equations for NSs.

References

- [1] C. Cutler and E. E. Flanagan. "Gravitational waves from merging compact binaries: How accurately can one extract the binary's parameters from the inspiral waveform?" In: *Physical Review D* 49.6 (Mar. 1994), pp. 2658–2697. ISSN: 0556-2821. doi: 10.1103/physrevd.49.2658.
- [2] R. G. Nascimento, K. Fricke, and F. A. Viana. "A tutorial on solving ordinary differential equations using Python and hybrid physics-informed neural network". In: *Engineering Applications of Artificial Intelligence* 96 (2020), p. 103996. ISSN: 0952-1976. doi: https://doi.org/10.1016/j.engappai.2020.103996.
- [3] M. Raissi, P. Perdikaris, and G. Karniadakis. "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations". In: *Journal of Computational Physics* 378 (2019), pp. 686–707. ISSN: 0021-9991. doi: https://doi.org/10.1016/j.jcp.2018.10.045.